

GLA UNIVERSITY



MINI PROJECT SYNOPSIS ON “MEDCONNECT — DOCTOR APPOINTMENT PLATFORM FOR UNIVERSITY HEALTH CENTRE”

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Submitted To

Faculty Name: Mr. Akash Kumar Choudhary
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DECLARATION

We hereby declare that the project entitled “MedConnect — Doctor Appointment Platform for University Health Centre” is our proposed mini project work developed as part of our academic curriculum.

The project is designed to provide a digital doctor appointment and queue-management solution for a college/university health centre. The proposed system will allow students to view doctors according to their medical categories, check available and booked appointment slots, reserve a suitable consultation slot and receive a digital appointment token.

We declare that the planning, design, development, testing and documentation of this project will be carried out by the following team members under the guidance of Mr. Akash Kumar Choudhary, Technical Trainer.

Team Members:

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INTRODUCTION

MedConnect is a web-based doctor appointment and queue-management platform designed specifically for the health centre of a college or university. In many educational institutions, students depend on a traditional walk-in system for medical consultations, which often results in long physical queues, overcrowding at the health centre and unpredictable waiting times. Students are frequently unaware of doctor availability, visiting hours or the number of patients already waiting, making the process inconvenient and time-consuming.

To address these challenges, MedConnect digitizes the entire appointment process by providing dedicated functionalities for students, doctors and administrators through a single, unified system. The platform enables the online scheduling and management of consultations, thereby reducing manual dependency and improving the overall efficiency of the health centre. The system is developed using the MERN Stack — MongoDB, Express.js, React.js and Node.js — to ensure a secure, scalable and responsive application.

ABOUT THE PROJECT

Through MedConnect, a student can select a medical category and view the list of doctors available under that category. Each doctor's profile displays relevant details such as specialization, room/cabin number, visiting days and OPD timings. After selecting a doctor and a preferred date, the student can view the consultation slots for that day, with available slots displayed in green and booked slots displayed in red. Each consultation slot is of fifteen minutes' duration. Once a suitable slot is selected and the appointment is confirmed, the system automatically generates a digital appointment/token number for the student. Students can also view their upcoming and previous appointments and cancel any of their own upcoming appointments when required.

Doctors can manage their daily appointment queue, update the status of each appointment and add brief medical remarks for reference. Administrators are responsible for managing doctor profiles, medical categories and overall appointment-related information within the system. The platform currently supports the following medical categories: General Physician, Gynecologist, Dentist, Orthopedic, Mental Health Counselor, Physiotherapist, Ophthalmologist and ENT Specialist.

Project Title

“MedConnect — Doctor Appointment Platform for University Health Centre using MERN Stack”

PRIMARY REASON TO CHOOSE THIS PROJECT

University students frequently require medical consultations for common health issues, but the traditional walk-in system can result in long queues and unnecessary waiting at the campus health centre.

A student may have to physically visit the health centre only to find that the preferred doctor is unavailable or that all consultation slots are already occupied. This wastes students' time and can create overcrowding during peak OPD hours.

Therefore, MedConnect is proposed as a digital solution that allows students to know the doctor, category, date, available slots and booked slots before visiting the health centre.

The major reasons for choosing this project are:

- Reduction of physical queues.
- Reduction in student waiting time.
- Easy identification of doctors according to specialization.
- Real-time visibility of available and booked slots.
- Convenient advance appointment booking.
- Prevention of duplicate appointment booking.
- Digital appointment/token generation.
- Better organization of doctor queues and appointment records.
- Reduced dependency on manual registers.

THE MAIN OBJECTIVE OF THE PROJECT

The main objective of MedConnect is to develop a secure, centralized and user-friendly doctor appointment platform for a university health centre. The system aims to:

1. Allow students to easily discover doctors according to medical specialization.
2. Display doctors' room/cabin numbers and OPD timings.
3. Provide date-wise appointment availability.
4. Display available and booked consultation slots.
5. Allow students to reserve a suitable consultation slot.
6. Generate a digital appointment/token number after successful booking.
7. Prevent two students from booking the same doctor and time slot.
8. Allow students to view upcoming and previous appointments.

SCOPE OF THE PROJECT

The scope of MedConnect covers the digital management of doctor appointments and queues within a college/university health centre.

Student Scope

Students will be able to:

- Register and securely log in.
- Browse medical categories.
- View doctors under each category.
- View doctor specialization and room number.
- View OPD/visiting timings.
- Select a consultation date.
- View available and booked slots.
- Book an available slot.
- Receive a digital token.
- View upcoming appointments.
- View appointment history.
- Cancel eligible upcoming appointments.

Doctor Scope

Doctors will be able to:

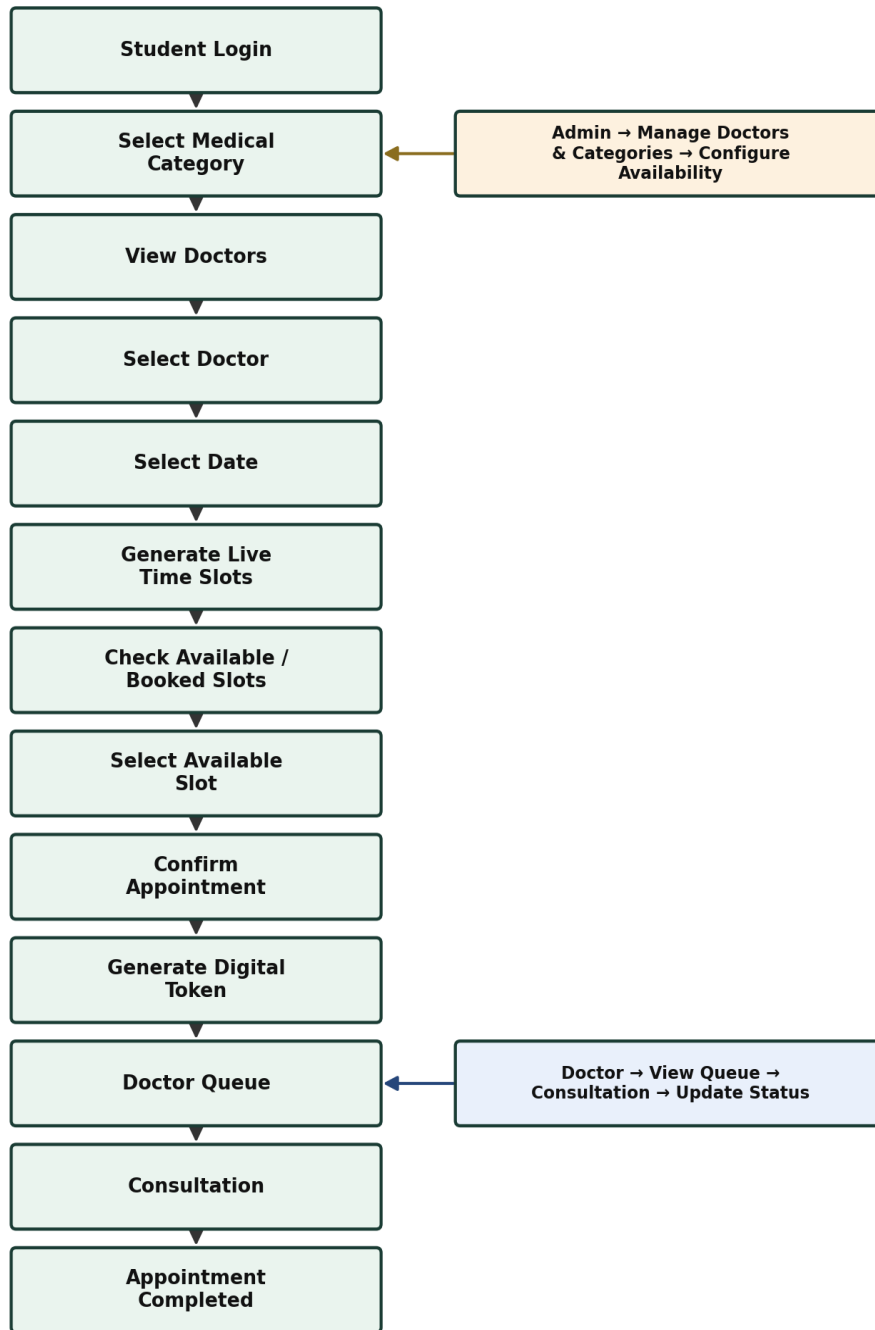
- View their doctor profile.
- Configure working/visiting hours.
- View their daily patient queue.
- View booked appointments.
- Mark appointments as completed or cancelled.
- Add brief medical remarks.

Admin Scope

Administrators will be able to:

- Add and manage doctor profiles.
- Manage medical categories.
- Configure doctor availability.
- View appointment information.
- Monitor basic queue/appointment statistics.

WORKING METHODOLOGY OF THE PROJECT



DETAILS ABOUT THE HARDWARE AND THE SOFTWARE

System Requirements

Supported Operating System:

- Windows 10 / Windows 11
- Linux
- macOS

Software Required:

- Visual Studio Code
- Node.js
- npm
- MongoDB
- Git and GitHub
- Modern Web Browser
- Postman
- React.js
- Express.js

Hardware Requirements

A basic computer system is sufficient for development and deployment testing. Recommended minimum configuration:

- Processor: Dual Core or better
- RAM: 4 GB minimum, 8 GB recommended
- Storage: At least 10 GB free space
- Stable Internet connection
- Keyboard and Mouse
- Monitor/Display

As MedConnect is a web-based MERN application, no Android Studio, IntelliJ or Eclipse-specific hardware requirements are applicable.

Listing Out Testing Technology

- Postman — API and backend endpoint testing.
- Browser Developer Tools — frontend debugging.
- Manual Functional Testing — login, doctor search, slot selection and appointment workflows.
- Boundary and validation testing — checking invalid dates, unavailable slots and incorrect input.
- Concurrency testing — verifying that two students cannot successfully book the same doctor/date/time slot.

FRONTEND AND BACKEND

Frontend

The frontend of MedConnect will be developed using the following technologies:

- HTML
- CSS
- JavaScript
- React.js
- React Router
- Tailwind CSS

The frontend will include login/register pages, a student dashboard, doctor category filter, doctor listing and detail views, an interactive time-slot grid, an appointment booking flow, digital token confirmation, appointment history, a doctor dashboard and an admin dashboard.

Backend

The backend of MedConnect will be developed using the following technologies:

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT (JSON Web Token) Authentication
- bcryptjs
- REST APIs

The backend will manage authentication, doctors, schedules, slots, appointments, tokens and role-based user access.

MODULE DESCRIPTION

1. Student Module

The Student Module allows university students to register/login, browse medical categories, search doctors, view doctor availability, select appointment slots, book appointments, receive digital tokens and manage appointment history.

2. Doctor Module

The Doctor Module provides doctors with a dedicated dashboard to manage their availability, view daily appointment queues, check patient details, update appointment status and add brief medical remarks.

3. Admin Module

The Admin Module allows health-centre administrators to onboard doctors, manage doctor categories, configure doctor information and monitor appointment/queue activity.

4. Doctor & Category Module

This module maintains doctor information such as:

- Doctor name
- Specialization
- Room/Cabin number
- Bio
- Visiting days
- OPD timings
- Slot duration

5. Slot Management Module

This module dynamically generates consultation slots according to the doctor's working hours. Example:

- 10:00–10:15 → Available
- 10:15–10:30 → Booked
- 10:30–10:45 → Available
- 10:45–11:00 → Available

6. Appointment & Token Module

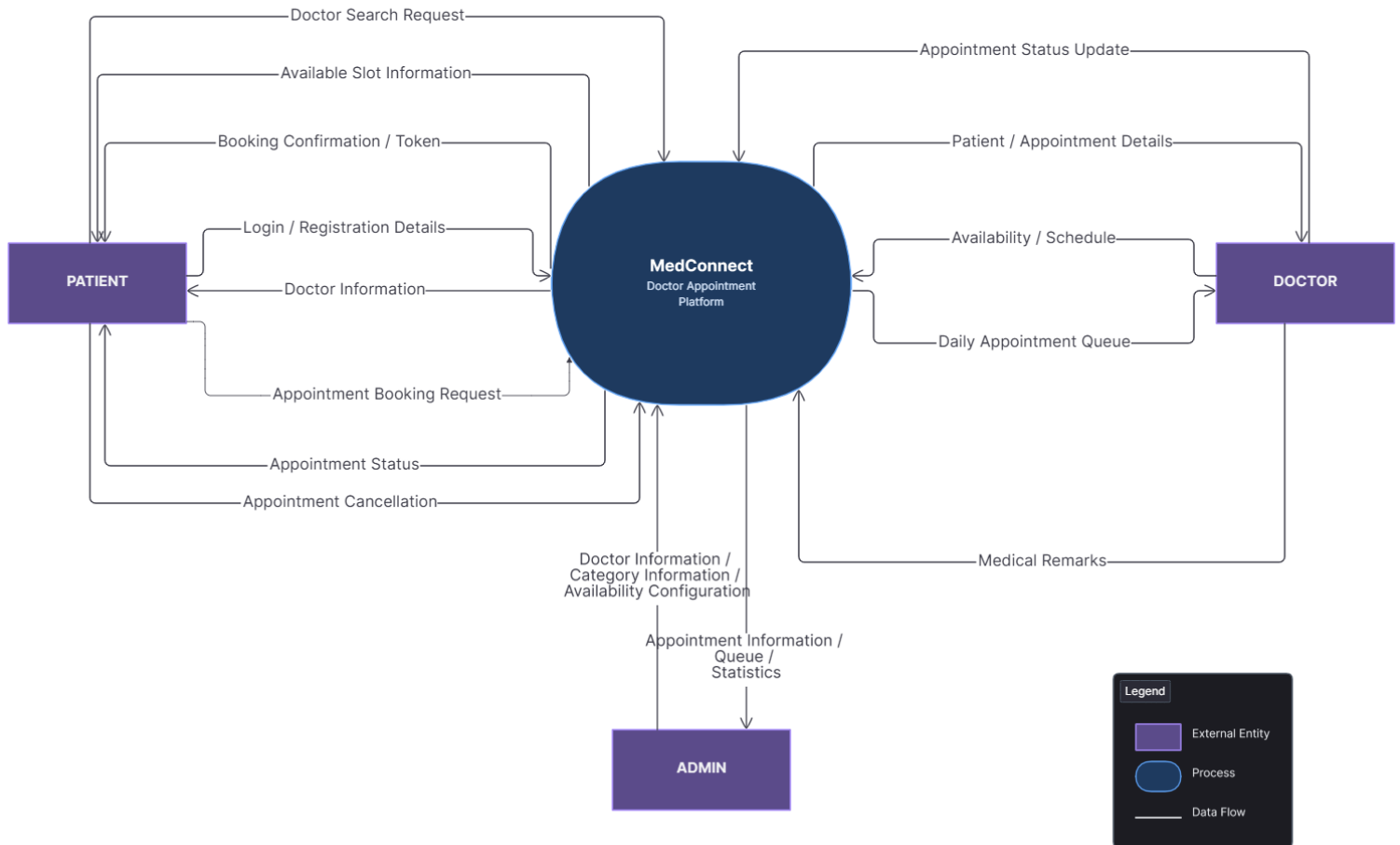
After a student selects an available slot, the system validates the booking and generates a digital token such as “TOKEN #07”. The system also prevents duplicate bookings of the same doctor, date and time slot.

DATA FLOW DIAGRAMS

A Data Flow Diagram (DFD) illustrates how data is processed by a system in terms of inputs and outputs. Its focus is on the flow of information — where data comes from, where it goes and how it gets stored. The following diagrams present the 0 Level, 1 Level and 2 Level DFDs of MedConnect.

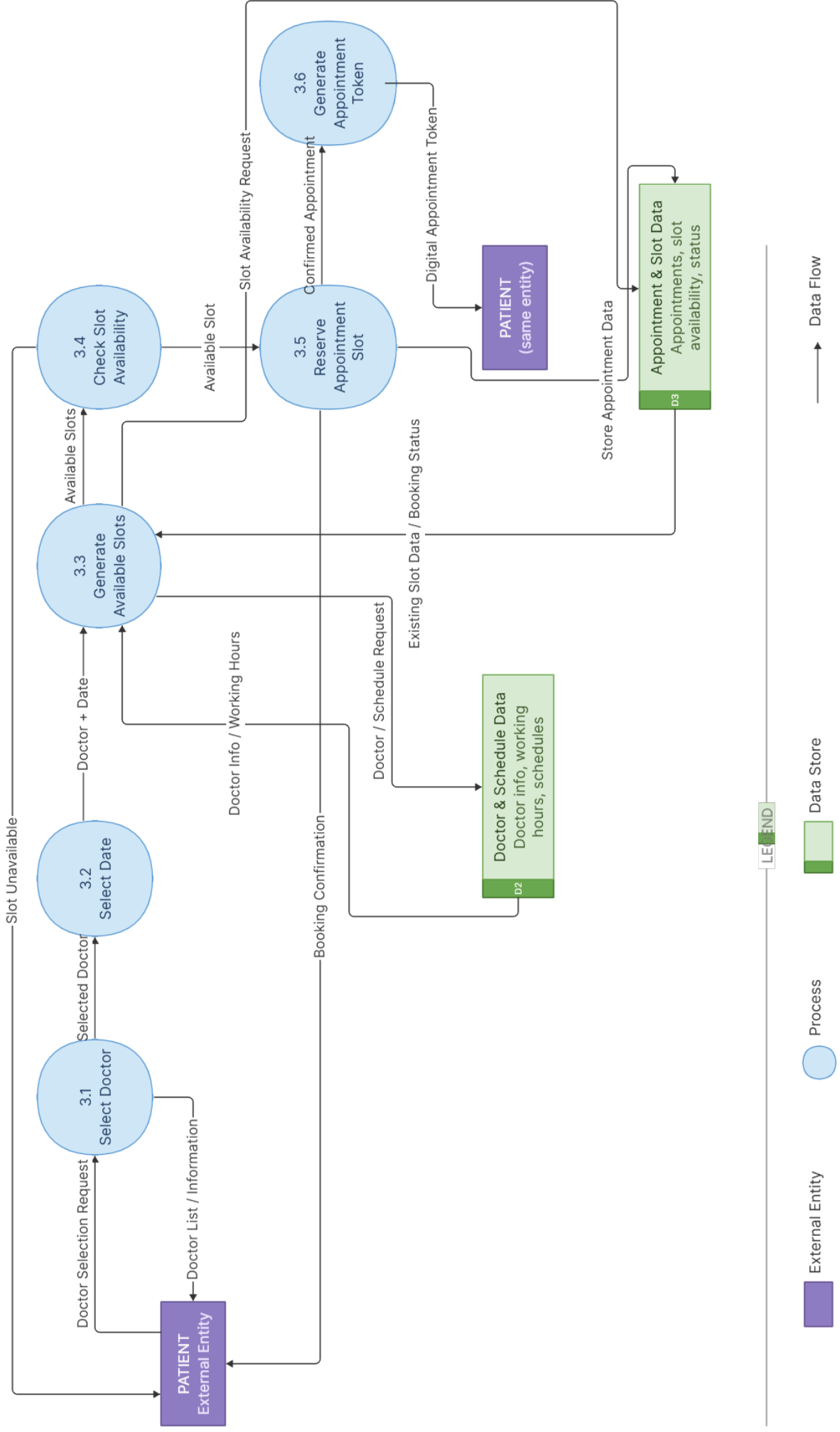
LEVEL 0 DFD – CONTEXT DIAGRAM

MedConnect — Doctor Appointment Platform



LEVEL 2 DFD – SLOT & APPOINTMENT MANAGEMENT

MedConnect — Doctor Appointment Platform



LE END

BIBLIOGRAPHY

The project development will refer to official documentation and learning resources related to the MERN stack, web development and database management.

References

1. React Documentation

<https://react.dev/>

Used for React components, hooks and frontend development.

2. Node.js Documentation

<https://nodejs.org/docs/latest/api/>

Used for backend runtime and server-side JavaScript development.

3. Express.js Documentation

<https://expressjs.com/>

Used for REST APIs, routing and middleware implementation.

4. MongoDB Documentation

<https://www.mongodb.com/docs/>

Used for database design, MongoDB operations and data management.